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Built only from: **Artificial Intelligence (Probabilistic Reasoning and Bayesian Networks)** — https://nptel.ac.in/courses/106105077

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Covers 7 of 7 syllabus sub-topics of this unit.

Acting under uncertainty

Definition

Acting under uncertainty refers to real-world agents that must act in environments characterized by partial observability, non-deterministic action outcomes, and noisy sensor data. Pure logical reasoning fails because of qualification problems, laziness, and theoretical ignorance. Probability theory provides the formal framework for quantifying degrees of belief under incomplete information.

Explanation

Prior probability P(A) represents the degree of belief in proposition A in the absence of any other information. Conditional probability P(A|B) denotes the probability of A given evidence B, formally defined as P(A|B)=(P(A and B) / P(B)) when P(B))0. Kolmogorov's axioms establish: (1) For any proposition A, 0(= P(A)(= 1. (2) P(True)=1 and P(False)=0. (3) For mutually exclusive propositions A and B, P(A or B)=P(A)+P(B). A full joint probability distribution assigns a probability to every possible atomic event in the state space. While it completely specifies uncertainty and answers any probabilistic query by marginalization, its size scales exponentially as O(d^n) for n variables with domain size d, making direct storage and computation intractable.

Algorithm / Steps

1. Product rule: P(A and B)=P(A|B)x P(B)=P(B|A)x P(A).
2. Chain rule: P(X_1,X_2,...,X_n)=Product_i=1^nP(X_i|X_1,...,X_i-1).

Key points

- Real-world agents operate in environments with partial observability, non-deterministic actions, and noisy sensors.
- Pure logical reasoning fails due to qualification problems, laziness, and theoretical ignorance.
- Probability theory quantifies degrees of belief using prior and conditional probabilities.
- Kolmogorov's axioms define the properties of probability, and a full joint distribution fully specifies uncertainty but is intractable because its size grows exponentially.
- Product rule and chain rule decompose joint probabilities into simpler components.

Bayesian inference

Definition

Bayesian inference is the process of updating beliefs about a hypothesis or variable using Bayes' Rule, which relates prior beliefs, likelihood of evidence, and posterior beliefs. It provides a formal framework for reasoning under uncertainty.,

Explanation

Bayes' Rule expresses the posterior probability of a cause given observed evidence as P(Cause | Effect) = (P(Effect | Cause) x P(Cause) / P(Effect)). Here, P(Cause) is the prior, P(Effect | Cause) the likelihood, P(Effect) the marginal evidence (normalizing constant), and the result is the posterior probability. This rule enables diagnostic reasoning, inferring underlying causes from observed symptoms, and is often easier to acquire from experts or data. In Bayesian networks, inference tasks compute posterior distributions P(X | E=e) by summing over hidden variables and can be performed exactly via variable elimination or approximately via sampling.,,

Algorithm / Steps

1. Compute the joint probability of the query variable, evidence, and hidden variables: P(X, E, Y).
2. Sum over all hidden variables Y to obtain P(X, E): Sum_y P(X, E, Y=y).
3. Normalize with the evidence probability alpha = 1/P(E) to get the posterior: P(X | E) = alpha x P(X, E).
4. Apply variable elimination to push summations inward across factors and cache intermediate products, reducing computational cost.

Key points

- Bayes' Rule updates prior beliefs with evidence to produce posterior probabilities.
- The prior, likelihood, and evidence together determine the posterior.
- Inference in Bayesian networks involves computing P(X | E) by summing over hidden variables.
- Exact inference can be performed by variable elimination, whose complexity depends on the network's tree width.
- When exact inference is infeasible, approximate methods such as stochastic simulation and sampling are used.

naïve bayes models. Probabilistic reasoning

The Naive Bayes classifier simplifies Bayesian inference by making a strong conditional independence assumption: given the target class variable C, all feature attributes X1, X2, ..., Xk are conditionally independent of each other. Formally: P(X1, X2, ..., Xk | C) = Product from j=1 to k of P(Xj | C).

Given observed evidence e = (x1, x2, ..., xk), the agent predicts the class label C that maximizes the posterior probability: C_optimal = argmax_c P(C=c) * Product from j=1 to k of P(Xj=xj | C=c).

Despite its naive assumption of attribute independence, Naive Bayes performs remarkably well in practice, particularly in text classification, document filtering, and medical diagnosis. It requires only linear time O(n * k) for training and evaluation and scales efficiently to high-dimensional feature spaces.

Bayesian networks

Definition

A Bayesian Network is a directed acyclic graph that represents direct probabilistic influence between variables, with each node associated with a conditional probability table (CPT) that quantifies P(X_i | Parents(X_i)).

Explanation

The joint distribution over all variables is compactly encoded as the product of the local conditional distributions: P(X_1, X_2, ..., X_n) = Product_i=1^n P(X_i | Parents(X_i)). This representation yields exponential savings in parameter storage when each node has at most k parents, scaling as O(n * 2^k) instead of O(2^n). Conditional independence is captured by the Markov blanket of a node (its parents, children, and the other parents of its children), and d-separation provides a graph-theoretic test for independence given evidence. Exact inference computes posterior distributions P(X | E = e) by summing over hidden variables, which can be done efficiently by variable elimination; the complexity depends on the tree width of the induced graph and is polynomial for singly connected networks but NP-hard in general. When exact inference is infeasible, approximate inference uses stochastic simulation and sampling methods.

Algorithm / Steps

1. Compute the joint distribution as the product of all CPTs.
2. Perform variable elimination by pushing summations inward across factors and caching intermediate products.
3. Generate samples using prior sampling, rejection sampling, or likelihood weighting, weighting each sample by the likelihood of the evidence.
4. Apply Markov Chain Monte Carlo methods such as Gibbs sampling, which iteratively samples non-evidence variables conditioned on their Markov blanket.

Key points

- A Bayesian Network is a DAG with CPTs for each node.
- The joint distribution is the product of local conditionals, providing exponential savings.
- Conditional independence is defined by the Markov blanket and d-separation.
- Exact inference is polynomial in polytrees but NP-hard in general; approximate inference uses sampling.

exact inference in BN

Definition

Exact inference in Bayesian Networks refers to computing the posterior distribution of a query variable given evidence by summing over hidden variables. The inference task is to compute P(X | E = e) = alpha · P(X, E = e) = alpha · Sum_y P(X, E = e, Y = y) where X is the query variable, E = e is the observed evidence, Y denotes the hidden variables, and alpha is a normalizing constant.

Explanation

Variable elimination improves efficiency by using dynamic programming: it pushes summations inward across factors and caches intermediate factor products. This reduces the exponential cost of standard enumeration by reusing partial results. The time and space complexity of exact inference by variable elimination is governed by the tree width of the network's induced graph. Exact inference is polynomial in polytrees (singly connected networks) but NP-hard in general multiply connected networks, necessitating approximate methods when the network is large.

Algorithm / Steps

1. Identify the query variable X and the evidence set E = e.
2. Enumerate the hidden variables Y that are not observed.
3. Compute the joint distribution P(X, E = e, Y) by multiplying the relevant conditional probability tables.
4. Sum over all possible values of Y to obtain P(X, E = e).
5. Normalize the result to obtain P(X | E = e).

Key points

- Exact inference computes the posterior distribution of a query variable given evidence.
- Variable elimination uses dynamic programming to reduce computational cost.
- Complexity depends on the tree width; polynomial for polytrees, NP-hard otherwise.

approximate inference in BN

Definition

Approximate inference in Bayesian networks refers to methods that estimate posterior distributions when exact computation is infeasible.

Explanation

Exact inference is polynomial in polytrees but NP-hard in general multiply connected networks, so approximate methods are required. These methods generate large collections of random samples from the joint distribution to estimate posterior probabilities.

Key points

- Approximate inference is used when exact inference is computationally infeasible.
- It relies on stochastic simulation and sampling techniques.
- Large collections of random samples are generated from the joint distribution.
- Posterior probabilities are estimated from these samples.

causal networks.

Definition

Causal networks are directed acyclic graphs in which directed edges strictly mirror physical causal relationships, going from cause to effect.

Explanation

Because the edges represent true causal influence, constructing a causal network yields a sparser topology, requires fewer conditional probability tables, and guarantees intuitive conditional independence properties. Causal networks support reasoning from cause to effect (prediction), from effect to cause (diagnosis), and across common effects (explaining away). They distinguish between passive observation and active intervention, allowing evaluation of interventions and counterfactual queries.

Algorithm / Steps

1. Prior Sampling: generate events by sampling each variable in topological order conditioned on its parents' sampled values.
2. Rejection Sampling: sample complete events from the prior distribution and reject any sample inconsistent with the observed evidence.
3. Likelihood Weighting: fix evidence variables to their observed values and weight each sample by the likelihood of the evidence given its parents: w = Product P(evidence_i | parents(evidence_i)).
4. Markov Chain Monte Carlo (MCMC) and Gibbs Sampling: initialize variables randomly with evidence fixed, then repeatedly sample one non-evidence variable at a time conditioned on the current values of its Markov blanket; in the limit the fraction of time spent in each state converges to the true posterior distribution.

Key points

- Directed edges in a causal network represent physical causal relationships from cause to effect.
- Causal networks are sparser and require fewer CPTs, yielding intuitive conditional independence.
- They enable reasoning from cause to effect, effect to cause, and across common effects.
- Causal reasoning distinguishes passive observation from active intervention, supporting counterfactual queries.

- Approximate inference methods such as prior sampling, rejection sampling, likelihood weighting, and Gibbs sampling can be applied to causal networks.

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